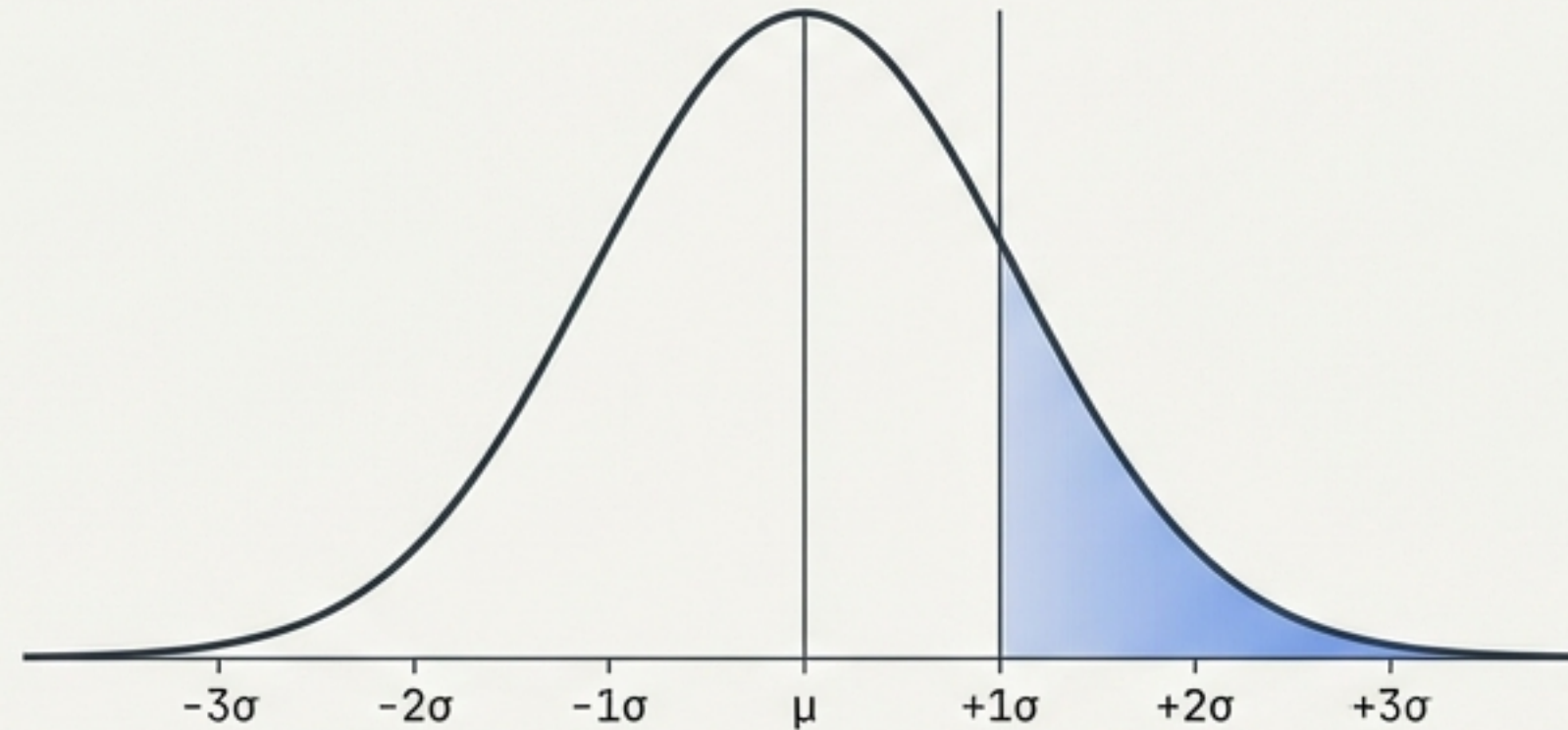


Mastering the T-Test

From Theory to Python Implementation: The Statistical Investigator's Handbook for One-Sample and Two-Sample Analysis.



STANDALONE REFERENCE GUIDE

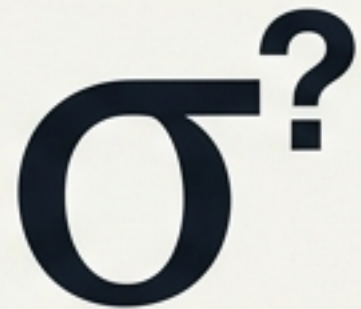
The Executive Summary: Uncovering Truth in Small Samples

Definition:

A statistical method used to determine if there is a significant difference between the means of two groups or against a standard.



**Small Sample
Size ($n < 30$)**

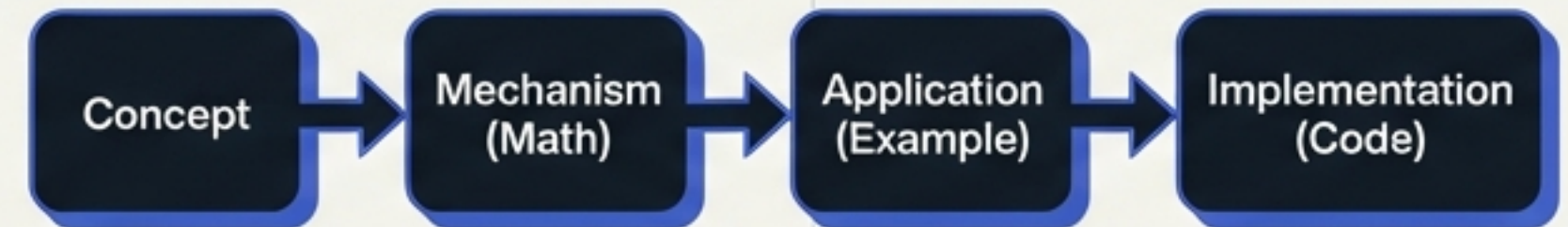


**Unknown
Population SD**

The Problem:

How do we distinguish signal from noise when we lack Big Data?

The Workflow:



Anatomy of the One-Sample Test

The diagram shows the t-test formula with labels for its components connected by lines:

- Sample Mean**
(The evidence collected) — points to $\bar{x}$
- Population Mean**
(The hypothesised standard) — points to μ
- Sample Standard Deviation**
(The variation within data) — points to s
- Sample Size**
(Must be < 30) — points to n
- Standard Error Adjuster** — points to the $\sqrt{}$ symbol

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

Degrees of freedom (df) = $n - 1$

Key Concept

The formula measures the ratio of the difference found to the variation expected. It is a signal-to-noise ratio.

Setting the Rules: Hypotheses and Tails

The Null Hypothesis (H_0)

$\mu = \mu_0$ (Status quo; no difference exists).

The Alternative Hypothesis (H_1)

A



Two-tailed: $\mu \neq \mu_0$

B



Left-tailed: $\mu < \mu_0$

C



Right-tailed: $\mu > \mu_0$

Checklist for Use

- Population SD known? ✗ No
- Sample size? ✓ Small ($n < 30$)
- Distribution? ✓ Approximately normal

Case Study 1: The Teacher's Claim (Two-Tailed)

The Question

A teacher claims the average score is 70.
A sample of 10 students has a mean of 69.7.
Is this difference real?

The Evidence

Sample Mean ($\bar{x}$) = 69.7

Standard Deviation (s) = 3.16

Sample Size (n) = 10

The Proof

Step 1: Calculate t

$$t = \frac{69.7 - 70}{3.16 / \sqrt{10}}$$
$$t \approx -0.30$$

Step 2: The Verdict



Fail to Reject H_0 . There is no significant difference.

Case Study 2: Battery Life Integrity (One-Tailed)

The Question

A company claims battery life is at least 12 hours ($H_0: \mu \geq 12$).

A sample of 8 devices averages 11.425 hours.

The Evidence

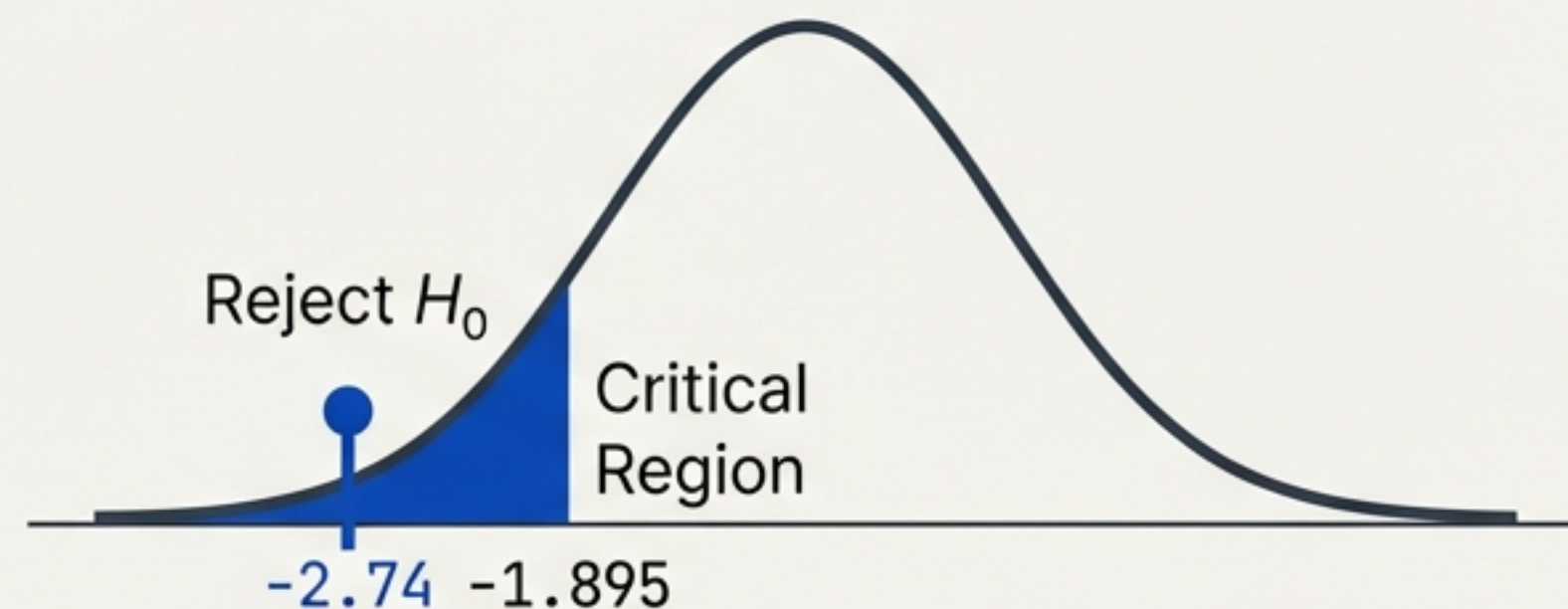
- $\bar{x} = 11.425$
- $s \approx 0.594$
- $n = 8$

The Proof

Step 1: Calculate t

$$t = \frac{11.425 - 12}{0.594 / \sqrt{8}} \approx -2.74$$

Step 2: The Verdict



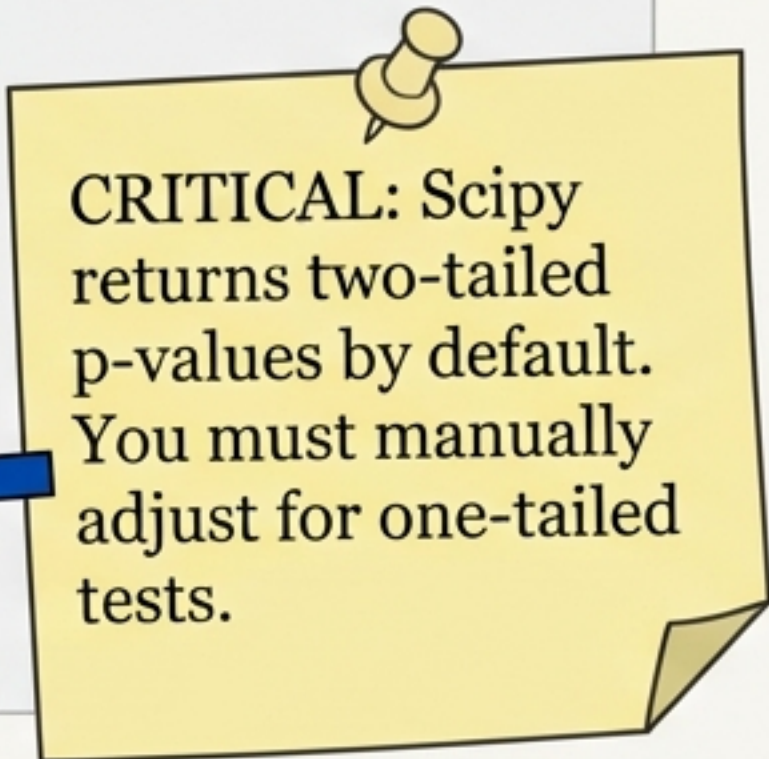
Reject H_0 . Battery life is significantly less than 12 hours.

The Python Lab: One-Sample Implementation

```
from scipy import stats

# Case 1: Two-Tailed Test
data1 = [68, 72, 65, 75, 70, 66, 74, 69, 71, 67]
t_stat, p_val = stats.ttest_1samp(data1, 70)
print(f'T-stat: {t_stat:.2f}, P-val: {p_val:.3f}') # Output: -0.30, 0.999
```

```
# Case 2: One-Tailed Test (Left)
data2 = [10.5, 11.2, 12.4, 11.8, 10.9, 12.1, 11.5, 11.0]
t_stat, p_val = stats.ttest_1samp(data2, 12)
# Adjust Scipy's default two-tailed p-value
p_one_tail = p_val / 2 if t_stat < 0 else 1 - (p_val/2)
print(f'Left-tailed P-val: {p_one_tail:.3f}') # Output: 0.021
```

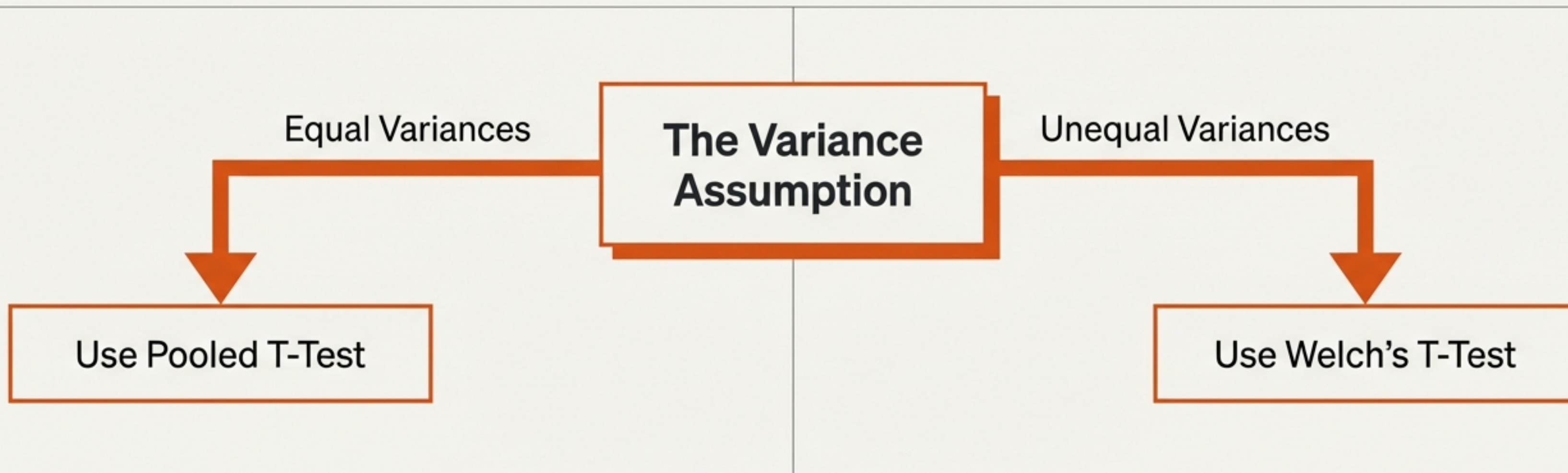


CRITICAL: Scipy returns two-tailed p-values by default. You must manually adjust for one-tailed tests.

The Two-Sample Test: Comparing Independent Groups

Definition: Determining whether two independent samples have significantly different means.

Common Use Cases: A/B Testing, Control vs. Treatment, User Group Analysis.



 Note: Both paths assume samples are from approximately normal distributions.

The Formula Fork: Pooled vs. Welch's

Path A: Pooled T-Test

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Where: $s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$.

Uses a shared “Pooled Standard Deviation”. Degrees of Freedom is simple: $df = n_1 + n_2 - 2$.

Path B: Welch's T-Test

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Keeps variances separate. Degrees of Freedom uses the complex Satterthwaite approximation.

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\left(\frac{s_1^2/n_1}{n_1-1}\right)^2 + \left(\frac{s_2^2/n_2}{n_2-1}\right)^2}$$

Case Study 3: Exam Scores (Equal Variance)

Scenario

Comparing Group A (n=5) vs. Group B (n=5).

Group A Mean: 86.6 | Group B Mean: 77.0

- **Assumption:** Variances are Equal.

Math Mechanics

Step 1: Calculate Pooled Variance (s_p)

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$
$$s_p \approx \sqrt{\frac{(4)(7.07)^2 + (4)(6.78)^2}{8}} \approx 6.93$$

Step 2: Calculate T-statistic

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{1/n_1 + 1/n_2}}$$
$$t = \frac{86.6 - 77.0}{6.93 \sqrt{1/5 + 1/5}} = \frac{9.6}{6.93 \cdot 0.632} \approx 2.19$$

Step 3: Degrees of Freedom

$$df = n_1 + n_2 - 2 = 5 + 5 - 2 = 8$$

 Critical Value (Two-tailed, $\alpha=0.05$) is ± 2.306 . Since $2.19 < 2.306$, we Fail to Reject H_0 .

Case Study 4: Group Differences (Unequal Variance)

Scenario

Group 1 (n=6, $\bar{x}$ =13.67, Var=2.27)

Group 2 (n=5, $\bar{x}$ =19.0, Var=2.5)

Observation: Unequal Sample Sizes and Variances

Math Mechanics (Welch's)

Step 1: The T-Calculation

$$t = \frac{13.67 - 19.0}{\sqrt{\frac{2.27}{6} + \frac{2.5}{5}}} \approx \frac{-5.33}{\sqrt{0.378 + 0.5}} = \frac{-5.33}{0.953} \approx -5.59$$

Step 2: Degrees of Freedom

df calculated via approximation ≈ 8.7



Critical Value (Two-tailed, $\alpha=0.05$) is ± 2.262 . Since -5.59 is far less than -2.262 , we **Reject H_0** . Significant difference found.

The Python Lab: Two-Sample Implementation

Scenario A: Equal Variance (Pooled) in Newsreader

```
stats.ttest_ind(groupA, groupB, equal_var=True)  
  
# Result matches Case Study 3:  $t = 2.19$ 
```

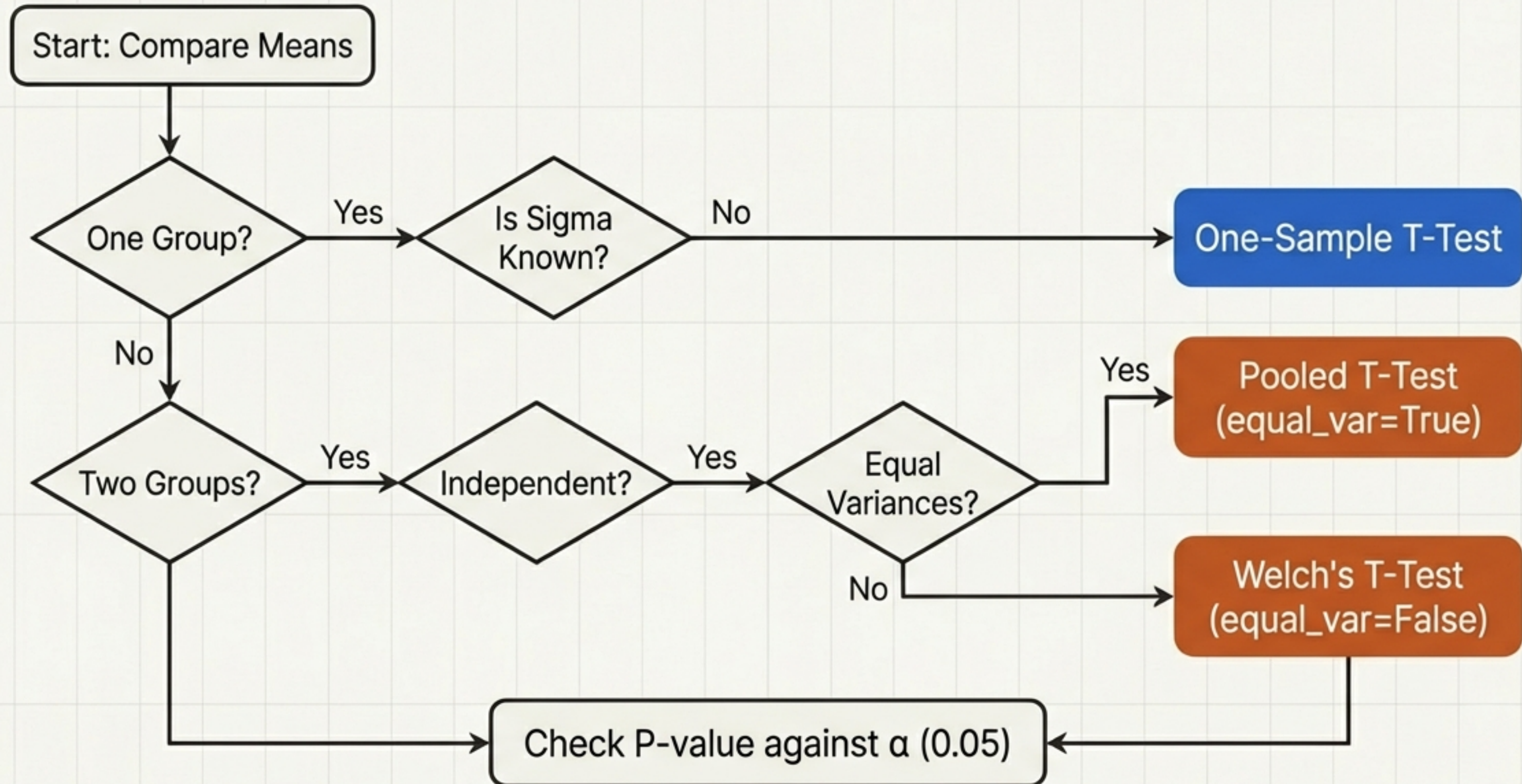
Scenario B: Unequal Variance (Welch's)

```
stats.ttest_ind(group1, group2, equal_var=False)  
  
# Result matches Case Study 4:  $t = -5.59$ 
```

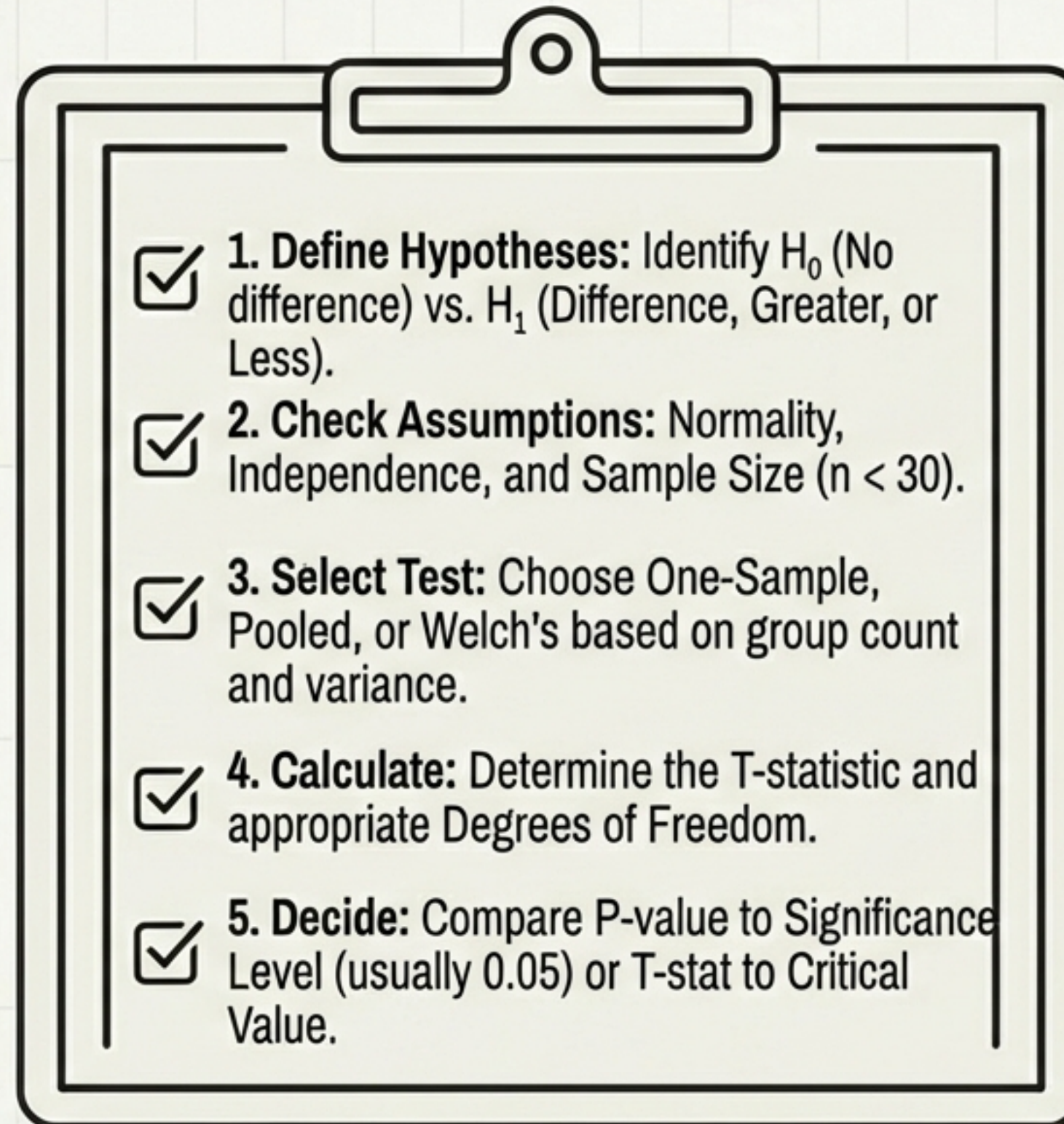


Pro-Tip: When in doubt, modern statistical practice often suggests defaulting to `equal_var=False` (Welch's t-test) as it is more robust against violations of homogeneity of variance.

The Statistical Decision Matrix



The Analyst's Checklist



Final Verdict

The T-Test is the bridge between a 'hunch' and 'proof'.

Whether testing a single product against a benchmark or comparing two distinct groups, the T-Test provides the mathematical rigour needed to navigate small datasets where the population standard deviation is unknown.

Call to Action: Use the Python snippets provided to automate this logic, but never lose sight of the assumptions that make the math valid.

